

Predicting Delays of Bay Area Flights

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The problem

Travelling is fun, but delays can easily ruin an otherwise great vacation. While airlines can tell us when delays come and might even know internally when they're coming, this often comes too late for passengers, who are left waiting at the gate or even on board for hours waiting, whether that's due to technical issues, weather, or anything else.

Setup

This project was created to predict delays in commercial flights departing from the San Francisco Bay Area. Several criteria were deliberately chosen:

- **Scope:** This scope of just Bay Area was chosen to reduce computational load, as I am unsure if my computer hardware can handle training from millions of rows in an effective timeframe. While rows could be filtered out for a training set, using more here is helpful because of how relatively infrequent delays are. The three main bay area airports used were OAK, SFO, and SJC. Data was used from Jan. 2023 to Dec. 2024. This was done to have full years (2025 is not complete yet), and to not go back far enough to where the pandemic still had major impacts on flights.
- **Feature Variables:** The design is such that input variables for prediction should be easily accessible to someone with an upcoming flight. At the same time, I also wanted to incorporate external features that not just in the single federal dataset, but also important factors I identified as a trained pilot, including weather conditions and aircraft type. Accordingly, these features are not easily ascertainable by an average person with a flight, but this data is public so a computer program could easily identify these for input given a flight number and date.
- **Target Variable:** The flight delay dataset has a continuous numeric variable for how many minutes flight departure was delayed. However, analysis of the data shown later shows it may be difficult to specifically predict how long a delay is. As such, the target variable will be a binary delay column, which is 1 if the delay column is 15+ minutes and 0 otherwise. Note this is focusing on flight departure, not arrival. Mechanical issues, weather, and other arrival airport conditions can easily alter an arriving aircraft's time. This focuses on departure time.
- **Datasets used:** Three datasets were used and mapped together to create an effective model:
 - o **[U.S. Bureau of Transportation Statistics Carrier On Time Performance](#):** This dataset contains a list of all domestic flights in the U.S. including carrier, aircraft, flight time, origin, destination, and delay time (if applicable). This was the main dataset with the target variables; the other two were mapped to entries in this one. The data is likely reliable as it comes from the U.S. government.
 - o **[Open-Meteo Historical Weather API](#):** This dataset was used to get weather and sunlight data for all 3 bay area airports (OAK, SFO, SJC) for every hour in 2023 and 2024. Data was mapped to entries by looking up weather for the given airport and time. Weather is a huge factor in aircraft departure times as there are strict limits on conditions aircraft can depart in.

- [Federal Aviation Administration Aircraft Registration Database](#): Aircraft Registration numbers from the delay dataset were referenced to retrieve aircraft information. Besides the obvious consideration of how reliable an aircraft type is, how big an aircraft is can impact whether or not it can fly in certain conditions (e.g. larger planes can handle more wind).

Data Pre-Processing

Getting all the data together and formatted was a laborious process. While I used AI to help with much of the code, it was still time consuming to identify gaps and make the correct directions for how to go about specific combination tasks. Some data imputation was also required. Many delay columns were missing, and since this was the target and a small percentage of total data, I decided to remove this. Many tail numbers and aircraft details were missing. This value was imputed using the most common aircraft type for each airline since these are generally fairly consistent. One challenge was that specific aircraft weight was not on the FAA dataset, so I used number of seats as a workaround.

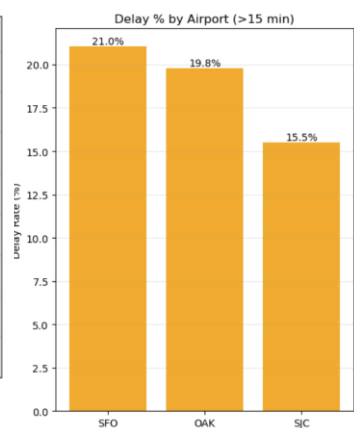
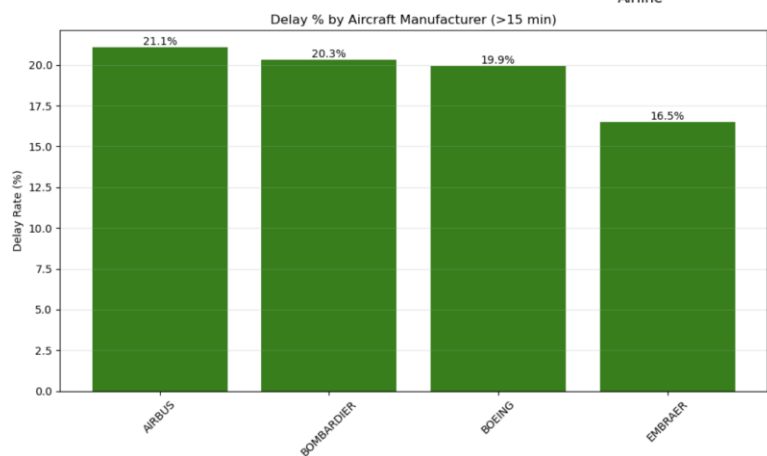
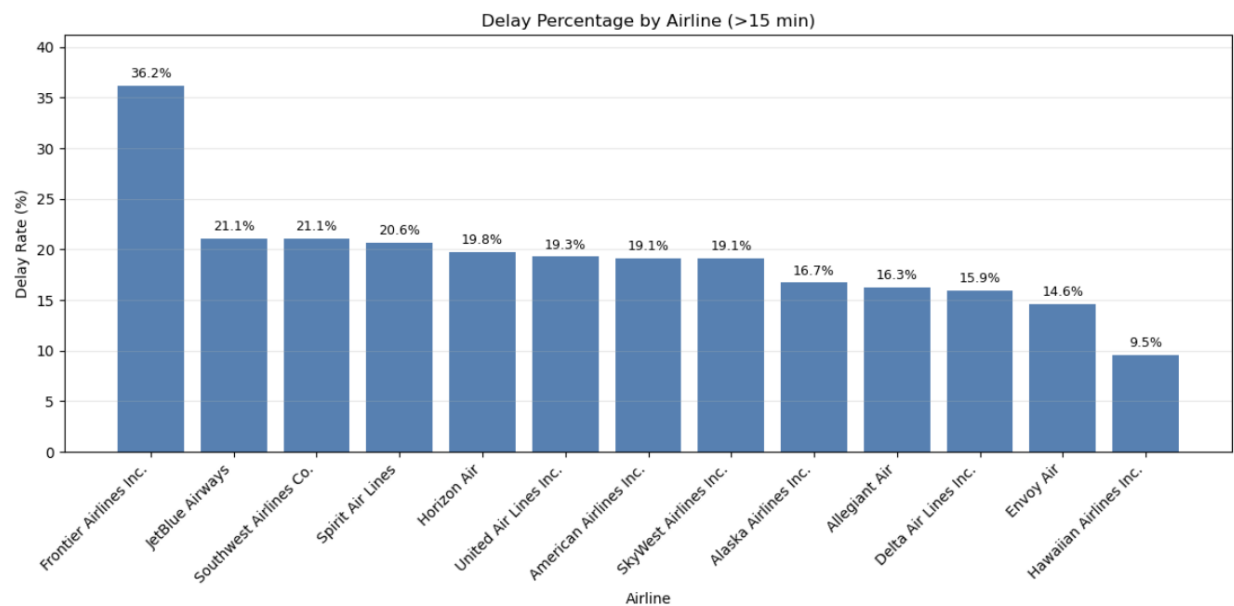
Data Analysis

Once data was collected, I had the following columns:

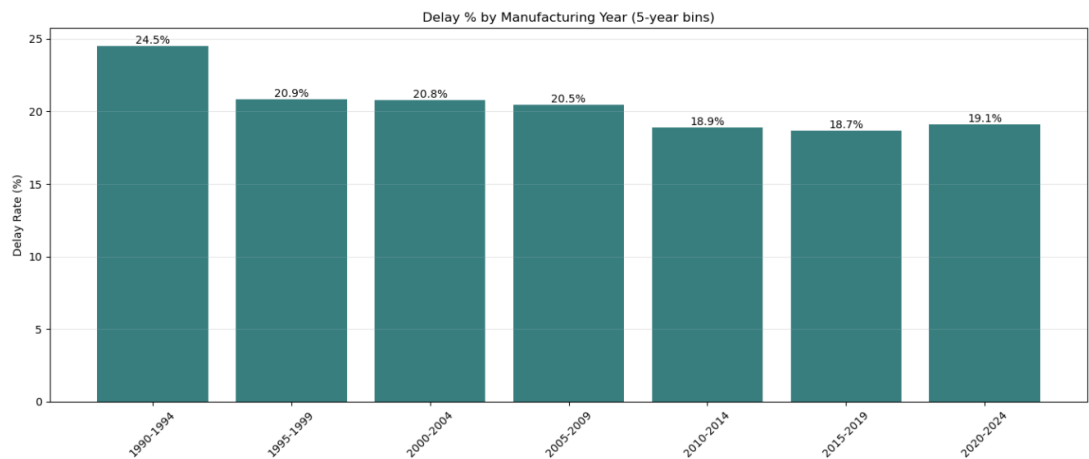
- **YEAR** (of scheduled departure time)
- **MONTH** (of scheduled departure time)
- **DAY_OF_MONTH** (of scheduled departure time)
- **DAY_OF_WEEK** (1 = Monday, 2 = Tues, etc.)
- **TAIL_NUM** (Similar to car VIN but for planes)
- **CRS_DEP_TIME** (Scheduled departure time)
- **DEP_DELAY_NEW** (delay in minutes)
- **temp_f** (temperature in Fahrenheit)
- **cloud_cover_low_pct** (Lower cloud cover since higher doesn't really matter)
- **precip_in** (precipitation in inches)
- **wind_speed_100m_mph** (wind speed in mph)
- **wind_gust_10m_mph** (wind gusting)
- **is_day** (is the sun out)
- **aircraft_mfr** (aircraft manufacturer)
- **aircraft_model** (aircraft model)
- **aircraft_num_seats** (number of seats in aircraft)
- **aircraft_year_mfr** (year aircraft was manufactured)
- **airport** (departure airport code)
- **DELAYED** (binary if flight has been delayed 15+ minutes)
- **CARRIER_NAME** (name of carrier)
- **DAY_OF_YEAR** (day of flight in year)

Using corr(), some correlations were found: departure time, wind gust, and wind speed have correlations with whether or not a flight will be delayed, but these are the strongest correlations and only at ~0.15

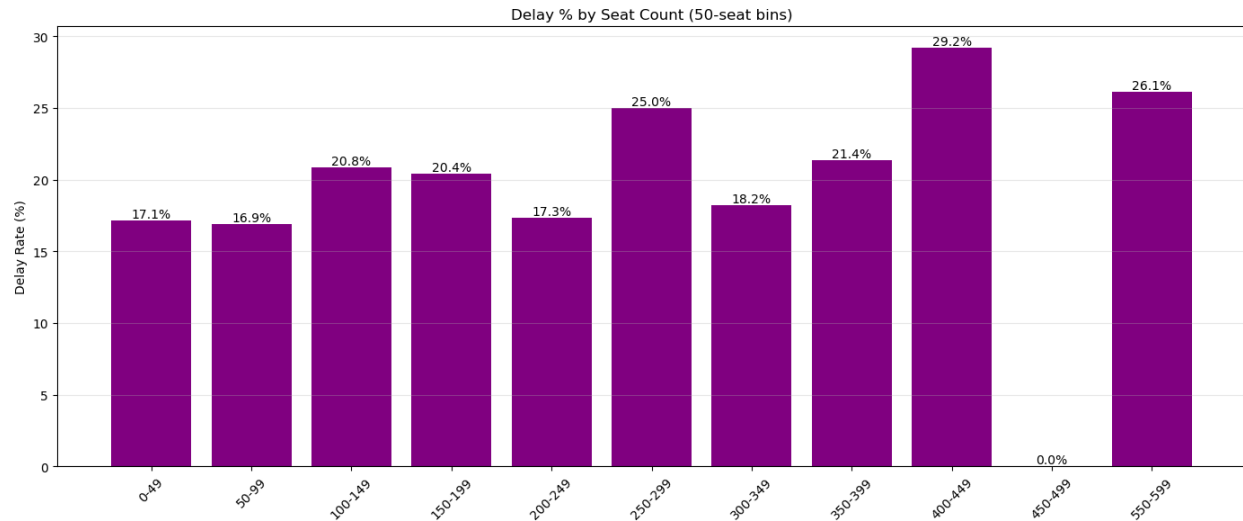
Avoiding delays means picking the right flight. We can use this dataset to analyze delays by airline, aircraft manufacturer and airport:



Relating to my earlier hypothesis, we can also see that older aircraft are more delayed:



However, interestingly, as seat number/aircraft size goes up, delays actually increase. This means that other factors like boarding issues, delays from previous legs, etc. are more of a factor than weather:



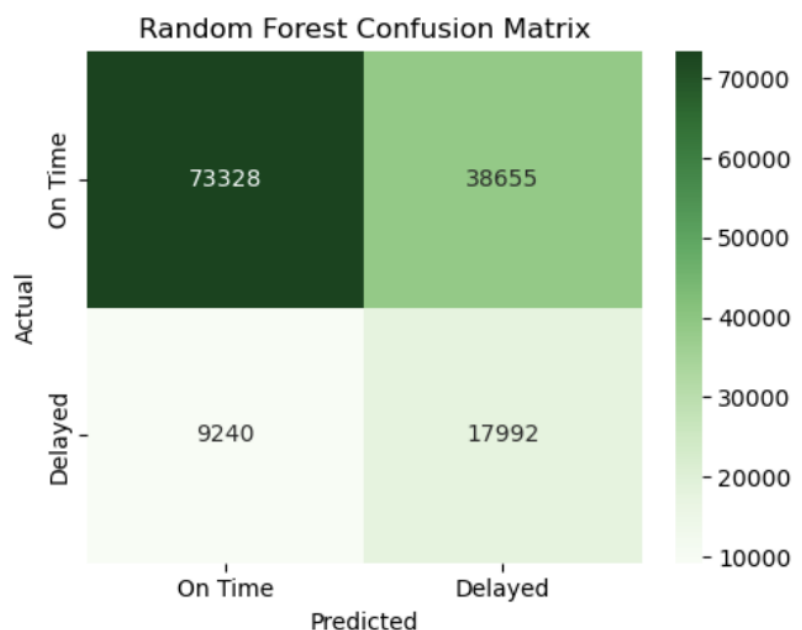
Machine Learning Model

I decided to use a random forest model here. There are several reasons. For one, this handles non linear factors better. Gradient descent would clearly fail, as some things are not purely linear. The clearest example is the day of the year feature. I added this column to account for high and low seasons in travel (e.g. around federal holidays and summer). By nature, much less of the flights in the dataset are delayed as compared to flights that were not delayed, and the dataset is also imbalanced. Additionally, research papers attempting a similar task to me also used random forests with better success than other classifiers. I also deliberately decided to not use a grouper like KNN because of performance reasons and how it would be difficult to manage having several categorically encoded and numeric variables factored in.

For features, I used all from above, except year, tail number, aircraft type, and delay in minutes (since delay was already a target variable). I one hot encoded categorical features, and used a 70/30 test-train split on the data.

Given the imbalance, I used class weighting during model training to give delayed flights more influence on the loss function. This was important because recall was the priority metric. A system that predicts “no delay” every time can achieve high accuracy, but is not very helpful.

Having chosen a random forest, I then tested various configuration, attempting to optimize for recall to get



the highest ratio of true delay and predicted delay out of all true delays, to avoid false positives. The best result I was able to achieve is shown, after one hot encoding category variables. I also used bootstrapping and class_weight to attempt to balance the dataset.

Having reviewed other academic research, I attempted to use a different form of random forest which we haven't learned about in class called Gradient Boosting Trees, or GBM using a framework called H2O. This functions like a random forest, but trees are iteratively created to lower error. The main disadvantage of GBM is they are prone to overfitting. I used the research paper's parameters for GBM and got the following results despite tuning:

Confusion Matrix (Act/Pred)			
	0	1	Error
0	41871	14158	0.2527
1	5853	7648	0.4335
Total	47724	21806	0.2878

This was unfortunate to see, and I clearly have a lot to learn. The main difference between my work and the paper was the research paper used arrival times instead of departures. This allowed them to use additional parameters like flight time, rather than using aircraft details. The research paper, as a result, was able to achieve 88% recall.

Interpretation of Results

Even in the best configuration, both models performed worse than I initially expected. They did achieve performance that is better than random guessing, but recall stayed relatively low compared to what would be needed for a truly useful decision tool for passengers. When I look at the confusion matrices for the Random Forest and GBM models, I saw a recurring pattern where the models are reasonably good at identifying flights that will be on time, but they miss many flights that are actually delayed. In other words, false negatives (predicting "on time" when the flight is delayed) occur more often than I would like, which directly conflicts with the goal of maximizing recall.

This is not very surprising when compared to the earlier correlation analysis. The strongest Pearson correlation with the binary delay label is departure time of day, and that correlation is only around 0.15. Other features such as wind speed, gusts, and precipitation have even smaller correlations. That means no single feature, and not even simple combinations of them, does a good job of separating delayed and non-delayed flights. The models are trying to learn a pattern that is weak and noisy, so they can only extract a limited signal.

The feature importance rankings from the tree based models reinforce this. Scheduled departure time is consistently the top feature. This makes sense as airports are busier at certain times of day, for example early morning banks of departures and evening peaks. More traffic can mean more conflicts on the ground and in the air, which increases the chance of pushing a flight back. Weather related features such as wind speed and gusts appear further down the importance list, especially for SFO, where crosswind limitations are a known operational constraint and go arounds are extremely common.

Delay rates differ by airline, which likely show differences in planning, buffer times, and how efficiently carriers recover from disruptions. For example, Alaskan and United have hubs in San Francisco. Additionally, budget airlines like Frontier and Southwest had high delay rates, which is because they have much shorter margins to save on costs. Older aircraft show higher delay rates on average which could

mean older airplanes are less safe or mechanically unreliable, but it should also be said that it may reflect the way airlines assign older aircraft to certain routes or the fact that older fleets are likely operated by the aforementioned budget airlines. The positive relationship between number of seats and probability of delay is also revealing. Larger airplanes usually serve busier routes and require more time for boarding, deplaning, cleaning, fueling, and baggage loading. Small disturbances in that chain can cause delays, and these effects propagate.

Many of the strongest causes of delays are operational and network based, not local, and these factors are not as present in the dataset. For example, whether the inbound aircraft arrived late, or whether there is a crew timing issue. None of these variables appear in my combined dataset outside of potentially being linked in carrier. The research paper that reached much higher recall used arrival based features such as complete flight times and delay propagation from earlier segments. That extra visibility into the network helps the model separate normal operations from cascading disruption. My model, by comparison, only sees local conditions at the departure airport plus some static aircraft characteristics.

From a traveler's point of view, the models are still somewhat informative. The system can show flights that are at higher risk of delay, for example those at peak times from certain airports under gusty conditions, but every problematic flight can't be shown. The prediction is more of a "risk indicator" than a firm yes or no answer.

Overall, the interpretation of the results is that delay prediction is fundamentally a system level problem. With only local and publicly available data, we can detect some broad patterns, yet the dominant sources of variability live in airline operations, traffic management, and upstream schedule disruptions. The models are telling a consistent story: the available features matter, but they do not tell the whole story, and that gap directly limits model performance.

Conclusion

The main findings of this project are mixed. On the positive side, the models do capture real patterns in the data. Time of day, windy conditions, airport choice, airline, aircraft age, and seat count all show some sort of relationship with delay rates. Both Random Forest and Gradient Boosting classifiers outperform a naive baseline that always predicts "on time." From a practical standpoint, this means that a traveler could gain some advantage by consulting a model like this when choosing between several flights.

There are several clear pathways for improvement. On the data side, integrating additional public sources such as holiday calendars, airport traffic statistics, and historical arrival delay data would likely provide stronger correlations and recall.

Despite all this, I still think the project succeeded overall at painting a picture of what causes delays, and both data analysis and key feature extraction led to some great insights into what really causes airline delays, including avoiding certain times, bad weather, and poor carriers (especially Frontier).

Works Cited

Kiliç, K., & Sallan, J. M. (2023). Study of Delay Prediction in the US Airport Network. *Aerospace*, 10(4), 342. <https://www.mdpi.com/2226-4310/10/4/342>

